

Hayden Fuller
S26 Internetworking of things
Lab 5
04/03/26

P1)

1) because we want to make sure all packets are reliably received, throughput and latency aren't an important factor

2) pros: it's very low power for the clients and low packet overhead for low bandwidth networks, it reduces the number of connections from $m*n$ to $m+n$

cons: it relies fully on one central broker, it can struggle with larger files, and it has no built in security

3) anything with many low power clients, namely sensor networks, but also slightly more advanced communication say between PLCs in an industrial setting

P2)
1)

The image shows a code editor window with a Python script. The script implements a scanner that connects to a target host, sends a 'FULLER/UID' message, and receives a response. It then prints the response and continues to track the target.

```
C:\Users\Hayden\AppData\Local\Temp\scp2739\home\hayden\Desktop\Lab5\Lab5_Ptly (ZombieAttack\github.io) - Sublime Text (UNGATED)
File Edit Selection Find View Goto Tools Project Preferences Help
BUNDLE index.html scripts x | style.css x | discordSkip.txt x | VirtualMachineAccount.txt x | SDnotebook.txt x | First Name: Hayden x | filament.txt x | finalst.txt x | Lab5_Ptly x | Lab5_PCpy x | untitled x | dates.txt x | x | +
19 .....self.records = {}
20 .....
21 .....# The callback for when the client receives a CONNACK response from the server.
22 .....def on_connect(self, client, userdata, flags, rc):
23 .....    print("Connected to:", client_host, "port:", client_port)
24 .....    print("Flags:", flags, "returned code:", rc)
25 .....    client.subscribe("FULLER/UID", qos=0)
26 .....
27 .....# The callback for when a message is published.
28 .....def on_publish(self, client, userdata, mid):
29 .....    pass
30 .....
31 .....# The callback for when a message is received.
32 .....def on_message(self, client, userdata, msg):
33 .....    message = str(msg.payload.decode("utf-8"))
34 .....    if msg.topic == "FULLER/UID":
35 .....        serial_number, name, uid = message.split("#")
36 .....        if serial_number == self.pi_number:
37 .....            self.records[name] = uid
38 .....            print("{}'s check-in at scanner {} authenticated with uid: {}".format(name, serial_number, uid))
39 .....
40 .....def start_tracking(self):
41 .....    self.client.connect(
42 .....        THE_BROKER,
43 .....        port=1883,
44 .....        keepalive=60)
45 .....    self.client.loop_start()
46 .....    while True:
47 .....        id_, name = self.reader.read()
48 .....        if name != None:
49 .....            name = name.strip()
50 .....            timestamp = datetime.now(eastern).strftime("%Y-%m-%d %H:%M:%S")
51 .....            self.client.publish("FULLER/MFRC", payload="#".join([self.pi_number, name, timestamp]), qos=0, retain=False)
52 .....            time.sleep(3)
53 .....
54 .....def stop_tracking(self):
55 .....    self.client.loop_stop()
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```

The image shows a Visual Studio Code editor window with a Python script for an MQTT client. The script is named 'Lab5_PCPy.py' and is located in a project folder named 'Lab5_PCPy'. The script defines several functions:
 - `def __init__(self):`: Initializes the MQTT client with a specific host, port, and transport.
 - `def on_connect(self, client, userdata, flags, rc):`: Prints connection details and subscribes to the 'FULEH/MFRC' topic.
 - `def on_publish(self, client, userdata, mid):`: A simple pass-through function.
 - `def on_message(self, client, userdata, msg):`: Parses the received message to extract pi number, name, and timestamp, and stores them in a dictionary.
 - `def start_tracking(self):`: Connects to the MQTT broker and starts the loop.
 The interface includes a file explorer on the left, a code editor in the center, and a status bar at the bottom showing 'Line 1, Column 1' and 'Python'.

2)

```
(lab5) hayden@fullehrpi:~/Desktop/Lab5 $ python3 ../Lab1/RFIDtestWrite.py
/home/hayden/Desktop/Lab5/Lab5/lib/python3.13/site-packages/mfrc522/MFRC522.py:151: RuntimeWarning: This channel is already in
use, continuing anyway. Use GPIO.setwarnings(False) to disable warnings.
  GPIO.setup(pin_rst, GPIO.OUT)
New data:fulleh
Now place your tag to write
Written
(lab5) hayden@fullehrpi:~/Desktop/Lab5 $ python3 Lab5_PI.py
/home/hayden/Desktop/Lab5/Lab5_PI.py:13: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  self.client = mqtt.Client(client_id=CLIENT_ID, clean_session=True, userdata=None, protocol=mqtt.MQTTv311, transport="tcp")
Connected to test.mosquitto.org port: 1883
Flags: {'session present': 0} returned code: 0
fulleh's check-in at scanner 1 authenticated with uid: c8d7f4b06e784bef8365e618fb43733f

C:\Users\Hayden\Desktop\IOT\Lab5>python Lab5_PC.py
C:\Users\Hayden\Desktop\IOT\Lab5\Lab5_PC.py:8: DeprecationWarning: Callback API version 1 is deprecated, update to latest versi
on
  self.client = mqtt.Client(client_id=CLIENT_ID, clean_session=True, userdata=None, protocol=mqtt.MQTTv311, transport="tcp", ca
llback_api_version=mqtt.CallbackAPIVersion.VERSION1)
Connected to test.mosquitto.org port: 1883
Flags: {'session present': 0} returned code: 0
hiya checked in on scanner 1 at 2026-04-06 14:09:43
Hello Hayden byte0x000x000x000x000x000x000x000x00 checked in on scanner 1 at 2026-04-06 14:19:46
Hello Hayden byte0x000x000x000x000x000x000x000x00 checked in on scanner 1 at 2026-04-06 14:19:52
test checked in on scanner 1 at 2026-04-06 14:20:41
fulleh checked in on scanner 1 at 2026-04-06 14:21:17
this is nice checked in on scanner 1 at 2026-04-06 14:21:39
|
```

3)

Check-In

Scanner	Name	Timestamp	Assigned UID	Total: 4
1	hiya	2026-04-06 14:09:43	e4c06422b0bc4448861c9ded8fa863bf	
1	test	2026-04-06 14:20:41	a22dfa08f8e74076aa2c0c6333f016b8	
1	fulleh	2026-04-06 14:21:17	c8d7f4b06e784bef8365e618fb43733f	
1	this is nice	2026-04-06 14:21:39	515fec33486a4fcabce903769ce8fc13	

4)

mqtt					
No.	Time	Source	Destination	Protocol	Length Info
54	1.892364800	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
55	1.892785100	192.168.1.116	54.36.178.49	MQTT	123 Publish Message [FULLEH/UID]
57	1.895168000	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
66	1.974272100	54.36.178.49	192.168.1.116	MQTT	123 Publish Message [FULLEH/UID]
158	2.648215900	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
159	2.648747500	192.168.1.116	54.36.178.49	MQTT	123 Publish Message [FULLEH/UID]
160	2.651405800	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
183	2.797298200	54.36.178.49	192.168.1.116	MQTT	123 Publish Message [FULLEH/UID]
271	4.997568300	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
272	4.998176700	192.168.1.116	54.36.178.49	MQTT	123 Publish Message [FULLEH/UID]
273	5.000763200	54.36.178.49	192.168.1.116	MQTT	111 Publish Message [FULLEH/NFRC]
285	5.008076900	54.36.178.49	192.168.1.116	MQTT	123 Publish Message [FULLEH/UID]
Frame 54: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits) on interface \Device\NPF_{93099C48-5491-4E16-9980-735...}					
Ethernet II, Src: Ubiquiti_d9:0d:07 (78:a7:41:d9:0d:07), Dst: Intel_6d:96:fb (b4:0e:de:6d:96:fb)					
Destination: Intel_6d:96:fb (b4:0e:de:6d:96:fb)					
.....0..... = LG bit: Globally unique address (factory default)					
.....0..... = IG bit: Individual address (unicast)					
Source: Ubiquiti_d9:0d:07 (78:a7:41:d9:0d:07)					
Type: IP-v4 (0x0800)					
[Stream index: 0]					
Internet Protocol Version 4, Src: 54.36.178.49, Dst: 192.168.1.116					
Transmission Control Protocol, Src Port: 1883, Dst Port: 56724, Seq: 1, Ack: 1, Len: 57					
MQ Telemetry Transport Protocol, Publish Message					
[Expert Info (Note/Protocol): Unknown version (missing the CONNECT packet?)]					
[Unknown version (missing the CONNECT packet?)]					
[Severity level: Note]					
[Group: Protocol]					
Header Flags: 0x30, Message Type: Publish Message, QoS Level: At most once delivery (Fire and Forget)					
0011 = Message Type: Publish Message (3)					
.... 0... = DUP Flag: Not set					
.... 00.. = QoS level: At most once delivery (Fire and Forget) (0)					
.... ..0 = Retain: Not set					
Msg Len: 55					
Topic Length: 11					
Topic: FULLEH/NFRC					
Message: 312348656c6cf57697265736861726b46756c6c656823323032362d30342d30362031343a33393a3438					
0000 b4 0e de 6d 96 fb 78 a7 41 d9 0d 07 08 00 45 00 m p A E					
0010 00 61 f4 cb 40 00 30 06 ab 59 36 24 b2 31 c0 a8 a 0 Y6\$ 1					
0020 01 74 07 5b dd 94 cc cf 7f ad ec 72 90 ec 50 18 t [P					
0030 01 f6 5a 86 00 00 30 37 00 0b 46 55 4c 4c 45 48 7 07 --FULLEH					
0040 2f 4d 46 52 43 31 23 65 65 6c 6f 57 69 72 65 /NFRCI ellowire					
0050 73 68 61 72 6b 46 75 6c 6c 65 68 23 30 31 35 shankful lehn2026					
0060 2d 30 34 2d 30 36 20 31 34 3a 33 39 3a 34 38 -04-06 1 4:39:48					

Table:

MQTT Section	Bit Range (start from 0)	Which fields are in the section and what are their values? (provide screenshots as evidence)
Fixed Header	of full packet: byte 54, bits 432-439 of MQTT section: byte 0, bits 0-7	Message type: 0b0011, 0x3, Publish Message DUP flag: 0b0 not set, New Message QoS level: 0b00, Fire and Forget Retain: 0b0, not set, new subs will not receive
Variable Header	of full packet: bytes 55-68, bits 440-551 of MQTT section: bytes 1-14, bits 8-119	Msg Len: 0x37, 55 bytes Topic Len: 0x000b, 11 bytes Topic: FULLEH/MFRC note: payload=message-topic-2=55-11-2=42 bytes
Payload	of full packet: bytes 69-110, bits 552-887 of MQTT section: bytes 15-56, bits 120-455	Message: 1#HelloWiresharkFulleh#2026-04-06 4:39:48

Fixed Header:

```

▼ Header Flags: 0x30, Message Type: Publish Message, QoS Level: At most once delivery (Fire and Forget)
  0011 .... = Message Type: Publish Message (3)
  .... 0... = DUP Flag: Not set
  .... .00. = QoS Level: At most once delivery (Fire and Forget) (0)
  .... ...0 = Retain: Not set

```

Variable Header:

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Msg Len: 55
Topic Length: 11
Topic: FULLEH/MFRC

```

Payload:

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Message: 312348656c6c6f57697265736861726b46756c6c656823323032362d30342d30362031343a33393a3438
2f 4d 46 52 43 31 23 48 65 6c 6c 6f 57 69 72 65 /MFRC1#H elloWire
73 68 61 72 6b 46 75 6c 6c 65 68 23 32 30 32 36 sharkFul leh#2026
2d 30 34 2d 30 36 20 31 34 3a 33 39 3a 34 38 -04-06 1 4:39:48

```

5)

Message type, differentiates between all the message types, such as connect, publish, subscribe, unsubscribe, ping, etc, and all of their Acknowledgments

Duplicate flag, if no ack is received, packets may be resent with the duplicate flag set

QoS level, determines which quality of service will be used: fire and forget, delivered at least once, and delivered exactly once

Retain, determines if this packet will be retained by the broker so that a new subscriber can have this packet sent to them immediately

Message Length, communicates the length of the rest of the packet- 2 bytes of topic length, X bytes of topic, and Y bytes of payload

Topic Length, communicates the length of the topic that follows

Topic, the topic that this will be published to

Payload, the data being sent, usually including a timestamp

6)

54.36.178.49:1883

IP in IPV4, port in TCP

7)

check everything for typos.

If I had coded anything myself, debug it.

test that my internet is working at all

`ping www.google.com`

ping the broker

`ping test.mosquitto.org`

test with different more unique topic (swap FULLEH in files)

check DNS with

`nslookup test.mosquitto.org`

test TCP with

`nc -vz test.mosquitto.org 1883`

check that no ports are closed by the firewall

`netsh advfirewall show allprofiles`

see if it's possible to subscribe to a topic

`mosquitto_sub -h test.mosquitto.org -t "FULLEH/MFRC" -v`

test if I can publish a message

`mosquitto_pub -h test.mosquitto.org -t "FULLEH/MFRC" -m "test message"`

7)

after installing mosquitto as a broker, replace the broker with my local IP address, found with

`ipconfig`

192.168.1.116